THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which sequence correctly lists in increasing size: red blood cell (7 micrometres), bacterium (2 micrometres), virus (0.1 micrometre)? 1

(a) virus, bacterium, red blood cell **(b)** bacterium, virus, red blood cell
(c) red blood cell, bacterium, virus **(d)** virus, red blood cell, bacterium

2. Why can't antibiotic medicines treat the flu? 1

(a) Flu is caused by a protozoan **(b)** Flu is not a disease
(c) Antibiotics are weak **(d)** Flu is caused by a virus

3. Assertion (A): Hooke saw living cells in a slice of cork. Reason (R): Cork consists of dead cells with only the walls remaining. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. The sperm cell in humans has a tail. This suggests: 1

(a) it stores water **(b)** it makes food
(c) specialised for movement **(d)** it is a plant cell

5. The best explanation of why a leaf cell is rectangular is: 1

(a) it lacks a membrane **(b)** it has a nucleus
(c) it has a rigid cell wall **(d)** it has no cytoplasm

6. Assertion (A): Viruses are not considered to be living outside a host. Reason (R): They show no life activity until they enter a host cell. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. An ostrich egg is a single cell yet visible to the eye, whereas a bacterium is not. This shows that: 1

(a) bacteria are bigger than eggs **(b)** cell size varies widely among organisms
(c) all cells are equal in size **(d)** cells cannot be seen

8. A student labels a plant cell: wall (inner), membrane (outer). The error is: 1

(a) wall is missing **(b)** no error
(c) membrane is correct **(d)** wall is outside the membrane

9. **Assertion (A):** Animal cells have chloroplasts. 1
Reason (R): Chloroplasts contain chlorophyll and carry out photosynthesis.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** An onion peel is stained before observing it under a microscope. 1
Reason (R): A stain makes the transparent parts of the cell easier to see.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Distinguish between a pathogen and a decomposer using examples. 2

12. Why do bacteria in the human gut not harm us? Give one benefit. 2

13. Why are viruses not treated with antibiotics? 2

14. A mixture of yeast and sugar solution is kept in a flask with a balloon on top. The balloon inflates. Explain. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why is the cell sometimes called a 'factory'? Compare three parts with parts of a factory. 3

16. Why are the cells of a leaf and a root of the same plant not the same, even though they share the same genetic material? 3

17. Two students observed cheek cells: A saw a clear nucleus and B did not. Give reasons. 3

OR

A cell with thick walls and no nucleus is found in dead wood. Explain why it may lack contents. 3

18. Compare unicellular and multicellular organisms in terms of reproduction and lifespan (one point each) and one difference in complexity. 3

OR

Why are most cells microscopic? State two advantages of being small. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A teacher compares three living things: an amoeba in pond water, a human and a mango tree. She explains that an amoeba is a single cell that eats, moves and reproduces on its own, whereas a human is made of trillions of cells that work together. 5
(i) What is the basic structural and functional unit of life?
(ii) Give the term for an organism made of one cell.
(iii) Give the term for an organism made of many cells.
(iv) Why do multicellular organisms need cells of different kinds?

20. Explain how the structure of the cell fits the function in: 5
(i) nerve
(ii) RBC
(iii) root hair
(iv) muscle
(v) guard cell.

OR

Compare and contrast the roles of the cell wall, cell membrane and vacuole in a plant cell, one point each, and say why the wall does not control entry of materials. 5

END OF QUESTION PAPER — Total: 40 Marks